```
! pip install chromadb sentence-transformers

Collecting .48.9 (from chromadb)
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Collecting opente1emetry-sdk>=1.2.e (from chromadb)

import chromadb from sentence_transformers import
SentenceTransformer
```

```
model = SentenceTransformer( ' all -Mini LM- L6-v2 )
                                                                                : 94: UserWarning:
The secret •                 does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface : c.QLsetüngs/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets. warnings . warn
  (
modules.json: 100%                                                    349/349              32.9kB/s]

                                100%                                                  116/116              11.4kB/s]

README.md:        10.5k!?                883kB/s]

                            100%                                                53.0/53.0              5.12kB/s]

config.json: 100%                                                  612/612              55.5kB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HE _TOKEN to enable higher rate limits and faster downloads.
WARNING: huggingface_hub. utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
model.safetensors: 100%                                                90.9M/90.9M              454MB/s]

Loading weights: 100%                                                103/103              441.77it/s, Materializing param—pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all -MiniLM-L6-v2
Key                              Status

embeddings . position_ ids        UNEXPECTED
Notes :
 UNEXPECTED          : can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%                                                350/350              8.10kB/s]

vocab.txt:        232k!?              8.65MB/s] tokenizer.json:

                  466k!?              20.2MB/s]

                            100%                                                112/112              11.7kB/s]

config.json: 100%                                                  190/190              18.8kB/s]
```

```
documents =
    "Artificial Intelligence is the simulation of human intelligence. " , "Machine
    Learning is a subset of Al that learns from data.
    "Deep Learning uses neural networks. " ,
    "Natural Language processing works with text data. " ,
    "RAG combines retrieval and generation. " ,
    "Al is used in healthcare and finance. " ,
    "Supervised learning uses labeled data. " ,
    "Unsupervised learning finds patterns in data. " ,
    "Reinforcement learning learns through rewards. " ,
    "Neural networks are inspired by the human brain. "
```



```
embeddings = model. encode(documents) .tolist()
client = chromadb.C1ient()
collection = client. create
                                  _

for  i,  doc  in  enumerate(documents)  :
    collection.  add(  documents=  [doc]  ,
    embeddings=  [  embeddings  [i]],
    ids=[str(i)]
```

```
def retrieve(query, k=2):
    query_embedding = model . encode( [query]) .tolist()
    results          =          collection.          query(
        query_embeddings—query_embedding,
        n results=k


    return results[ ' documents '

def                          :
    retrieved docs = retrieve(query)

    answer = " " . join (retrieved_docs)
    return answer, retrieved docs
```

```
query = "What is Artificial Intelligence?"

answer , docs = rag_answer(query)

print( "Question : " , query)
print( "\nRetrieved Docs: " )
for d in docs:
    print ("     d)
                    Answer:
print(answer)
```

```
Question: What is Artificial Intelligence?
Retrieved Docs:
  Artificial Intelligence is the simulation of human intelligence.
  Machine Learning is a subset of Al that learns from data.
Final Answer:
Artificial Intelligence is the simulation of human intelligence. Machine Learning is a subset of Al that learns from data.
```

```
queries =
    "Explain machine learning",
    "What is deep learning?" ,
    "Explain NLP"


for q in queries:
    answer, docs = rag_answer(q)

        \n========================
        Question:", q)
        Answer:", answer)
print("Question:
print("Answer:                    ( "                    " )
```

```
========================
Question: Explain machine learning
Answer: Machine Learning is a subset of AI that learns from data. Supervised learning uses labeled data.

==================
Question: What is deep learning?
Answer: Deep Learning uses neural networks. Machine Learning is a subset of AI that learns from data.

========================
Question: Explain NLP
Answer: Natural Language Processing works with text data. Supervised learning uses labeled data.
```